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PCB DESIGN FOR SMART BEE HIVE - IOTRA LAB

What?

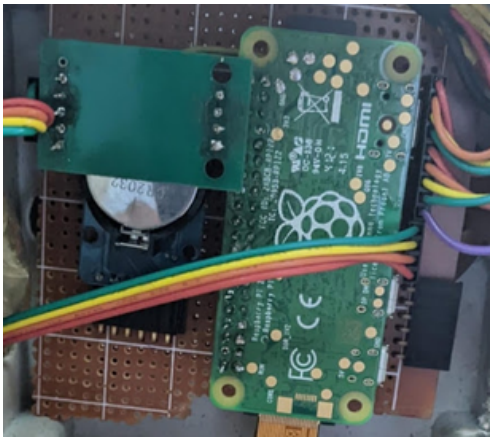
- Designed and implemented a comprehensive sensor integration system for smart beehive monitoring, incorporating **DHT22** temperature and humidity sensors, and weight sensors with a **Raspberry Pi** core.
- Transitioned from sensor integration with jumper wires on a Raspberry Pi to an initial prototype on a **perforated board**, and finally to a **professional-grade 2-layer PCB**, optimizing it for field deployment requirements.
- Created a **compact, weather-resistant** sensor arrangement that could withstand beehive environmental conditions while maintaining measurement accuracy.

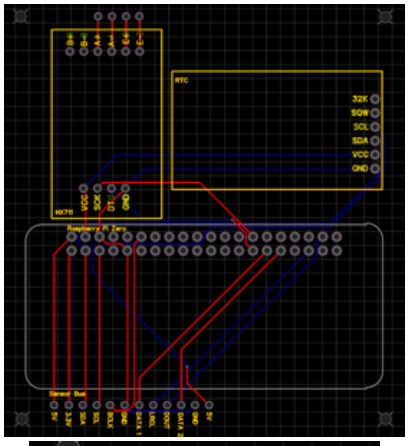
How?

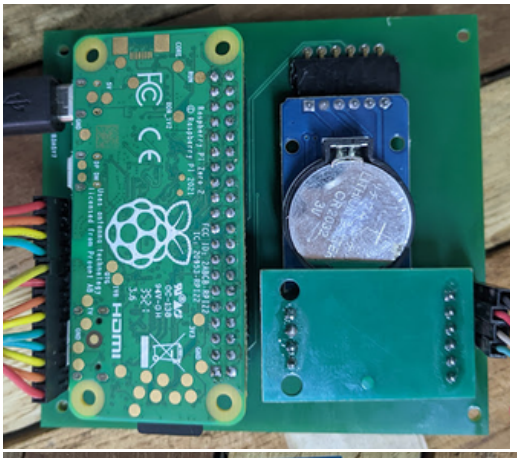
- Utilized **EasyEDA** for PCB design, implementing best practices for component placement and signal routing to minimize interference between analog and digital sections.
- Conducted thorough design validation including **DRC and ERC** before proceeding to manufacture.
- Collaborated with Chinese PCB manufacturer for production, managing specifications and quality requirements for **outdoor-capable** PCB.

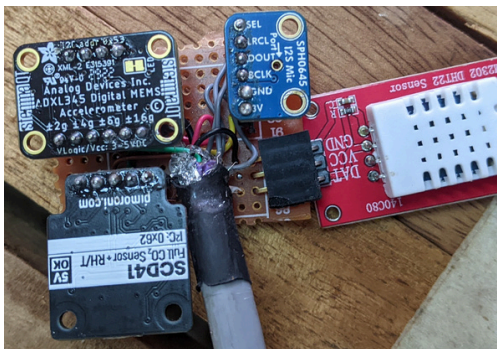
Results

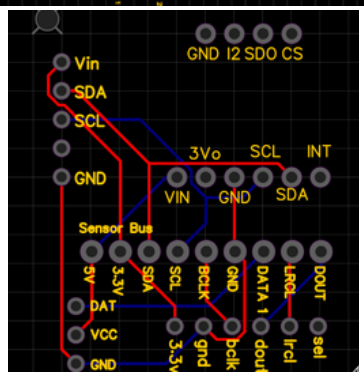
- Successfully transitioned from prototype to field-ready PCB design, eliminating **90%** of potential failure points from wire connections while achieving a compact form factor.
- Reduced assembly time from **4 hours** to **20 minutes** through elimination of manual wiring, enabling efficient scaling of production for multiple hive deployments.
- **Standardized** the sensor system assembly process with the PCB design, ensuring consistent performance across all deployed units and enabling reliable reproduction of the beehive monitoring setup.

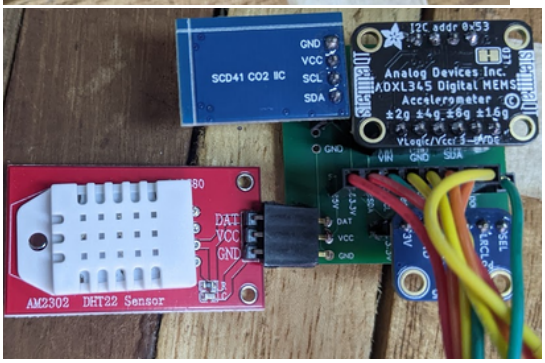












ACCELEROMETER FIRMWARE FOR SMART BEE HIVE - IOTRA LAB

What?

- Integrated **Adafruit ADXL345** accelerometer into existing sensor array, expanding system capabilities to detect and analyze bee colony activity patterns.
- Developed comprehensive firmware to capture, process, and analyze **vibrational data** from the beehive.
- Implemented sophisticated **signal processing algorithms** to distinguish between different types of colony behaviours.

How?

- Configured ADXL345 accelerometer with **1kH** sampling rate to capture the full spectrum of bee activity vibrations.
- Implemented **Fast Fourier Transform (FFT)** analysis to identify frequency signatures in the [X-Y Hz] range characteristic of specific bee behaviours.
- Developed a signal preprocessing algorithm that includes data acquisition at **10k samples per second** and **FFT processing** for **frequency domain analysis.**

Results

- Developed an **efficient signal processing pipeline** that processes vibration data while integrating with the existing hive monitoring system.
- Created data visualizations of **FFT output** that displayed frequency domain representations of hive vibrations.

